

CareScope : Hospital Patients , Services & Financial Analytics

Technology Stack:

Python | SQL / MySQL | Power BI | DAX

Project Focus:

Healthcare Data Cleaning, Database Analysis, Patient & Service Utilization, Financial Analysis and Interactive Business Intelligence

Prepared By:

SHASHWAT DINESH NAIK

Degree / Department:

BE ECE

Institution:

Dayananda Sagar College Of Engineering , Bengaluru

Executive Summary

The Healthcare Data Analysis & Business Intelligence project is an end-to-end data analytics project developed to transform raw healthcare data into a structured and interactive Business Intelligence solution.

The project follows a complete analytical workflow beginning with data cleaning and preparation using Python, followed by database development and SQL-based analysis, and finally integration with Microsoft Power BI for data modeling, DAX-based calculations, interactive visualization, and business reporting.

The healthcare dataset contains information related to patients, encounters, procedures, payers, organizations, claims, and insurance coverage. The analysis covers 27,891 healthcare encounters, 47,701 procedures, and 974 patients, allowing the project to examine healthcare activity from operational, patient, financial, and service perspectives.

The project was designed around four major analytical areas:

- Healthcare Overview — understanding the overall scale of healthcare activity, encounter distribution, claim costs, and historical trends.
- Patient Analysis — examining patient demographics, healthcare utilization, payer distribution, and the relationship between utilization and claim costs.
- Financial Analysis — analyzing claim costs, payer contribution, insurance coverage, and uncovered financial exposure.
- Services Analysis — examining procedure volume, procedure-to-encounter ratios, encounter classes, and typical visit duration.

Python was used to prepare the raw healthcare data, while SQL/MySQL provided the structured database layer and enabled analytical queries and aggregations. Power BI was then used as the Business Intelligence layer to create interactive dashboards, KPIs, filters, DAX measures, ratios, and visual analytics.

The final solution provides a connected analytical story:

Raw Healthcare Data → Python Cleaning → SQL/MySQL Database → SQL Analysis → Power BI → DAX & Visualization → Business Insights

The project therefore demonstrates the complete workflow of a practical Data Analyst / Business Intelligence project, rather than focusing only on dashboard creation. It provides stakeholders with a centralized way to understand patient utilization, encounter activity, healthcare procedures, claim costs, payer coverage, uncovered financial exposure, and service intensity.

Table of Contents

No.	Section	Page No.
1	Introduction	4
2	Business Problem and Objectives	5
3	Technology Stack, Dataset and Project Architecture	7
4	Data Preparation, SQL Analysis and BI Methodology	9
5	Dashboard 1 – Healthcare Overview	12
6	Dashboard 2 – Patient Analysis	14
7	Dashboard 3 – Financial Analysis	16
8	Dashboard 4 – Services Analysis	18
9	Key Business Insights	20
10	Conclusion	22

Introduction

CareScope is an end-to-end healthcare data analytics project designed to transform raw hospital data into meaningful business insights.

The project combines Python, SQL/MySQL, and Microsoft Power BI to analyze healthcare activity across patients, encounters, procedures, payers, claims, and insurance coverage.

Rather than performing analysis directly on raw data, the project follows a structured analytics workflow:

Raw Data → Python Cleaning → SQL Database → SQL Analysis → Power BI → Interactive Dashboards → Business Insights

The final Power BI solution consists of four analytical dashboards:

1. Healthcare Overview
2. Patient Analysis
3. Financial Analysis
4. Services Analysis

Each dashboard focuses on a different aspect of hospital activity while collectively providing a complete view of the healthcare dataset.

The purpose of the project is to demonstrate how raw healthcare data can be converted into a structured, business-ready analytical solution. The project covers data preparation, database organization, SQL analysis, KPI development, healthcare utilization analysis, patient analysis, financial analysis, service analysis, and interactive Business Intelligence reporting.

The overall objective is to demonstrate the complete lifecycle of a Data Analyst project, from raw data to business insights.

Business Problem and Objectives

Healthcare organizations generate large volumes of information through patient interactions, healthcare encounters, procedures, claims, insurance coverage, and payer transactions.

Raw healthcare data by itself does not directly answer important business questions. When different aspects of the data are analyzed separately, it becomes difficult to understand the relationship between patient utilization, healthcare services, procedures, claims, and financial exposure.

The central business problem addressed by CareScope is:

How can raw healthcare data be cleaned, structured, analyzed, and transformed into an interactive Business Intelligence solution that provides meaningful insights into patient utilization, healthcare services, claims, and financial performance?

The project objectives were to:

1. Clean and prepare raw healthcare data using Python.
2. Structure the cleaned data for database storage and analysis.
3. Use SQL to query and analyze healthcare information.
4. Connect the analytical data to Power BI.
5. Build business-oriented KPIs and analytical measures.
6. Analyze patient utilization and encounter patterns.
7. Analyze claim costs and payer coverage.
8. Identify high-volume healthcare procedures.
9. Analyze procedure intensity across encounter classes.
10. Present the findings through interactive dashboards.

The dashboards were designed to answer questions covering four major areas:

Healthcare Activity

- What is the overall scale of healthcare activity?
- How many patients, encounters, and procedures are recorded?
- Which encounter classes have the highest volume?
- How does encounter activity change over time?

Patient Utilization

- How frequently do patients use healthcare services?
- How does utilization differ across age groups?
- Which payers are associated with the patient population?

- Is higher utilization associated with higher claim costs?

Financial Performance

- Which payers account for the highest claim costs?
- How much of the claim amount remains uncovered?
- What is the distribution of claims by coverage status?
- Which encounter classes have the greatest claim burden?

Services

- Which procedures are performed most frequently?
- How does procedure volume vary by encounter class?
- What is the procedure-to-encounter ratio?
- What is the typical duration of a healthcare encounter?

Technology Stack, Dataset and Project Architecture

Technology Stack

Python : Data cleaning, preprocessing, transformation and preparation

MySQL : Database storage, joins, aggregations and analytical querying

Power BI : Data modeling, interactive visualization and dashboard development

DAX : KPIs, measures, ratios and analytical calculations

Python was used at the beginning of the pipeline to prepare the raw healthcare data.

SQL/MySQL was used to store the structured data and perform analytical queries.

Power BI was used as the final Business Intelligence layer for interactive dashboards and visual reporting.

This combination creates a realistic end-to-end Data Analyst workflow rather than a dashboard-only project.

Dataset

- Dataset Name: Hospital Patient Records
- Source: Maven Analytics
- Domain: Healthcare / Hospital Analytics
- Time Period: 2011–2022
- Key Entities: Patients, Encounters, Procedures, Payers, and Organizations

The dataset represents hospital patient records and healthcare activity across multiple years. It contains information about patients, their encounters, procedures performed, payers, and healthcare organizations.

The data supports analysis of patient utilization, healthcare services, procedures, claims, and payer coverage.

The dataset is structured across multiple related entities, making it suitable for database and SQL-based analysis.

It was used as the foundation for the Python → SQL/MySQL → Power BI end-to-end analytics workflow in the CareScope project.

The project uses a structured healthcare dataset containing information about healthcare activity across multiple years.

The dataset contains information related to:

- Patients
- Encounters
- Procedures
- Payers
- Organizations
- Claims
- Insurance coverage

Dataset Entities

Patients

Contains information about individuals receiving healthcare services, including attributes such as:

- Patient
- Gender
- Age
- Marital Status
- City

Encounters

Contains individual healthcare interactions and includes fields such as:

- Encounter ID
- Start
- Stop
- Patient
- Organization
- Payer
- Encounter Class

The Start and Stop timestamps were also used to derive encounter duration.

Procedures

Contains information about healthcare procedures performed during encounters and supports:

- Procedure volume analysis
- Top procedure analysis
- Procedure-to-encounter ratio
- Service analysis

Payers

Provides information required for:

- Patient-by-payer analysis
- Claim analysis
- Coverage analysis
- Uncovered claim analysis

Organizations

Contains healthcare organization information associated with encounters.

End-to-End Project Architecture

The complete workflow is:

RAW HEALTHCARE DATA



PYTHON DATA CLEANING



SQL / MYSQL DATABASE



SQL ANALYSIS & QUERIES



POWER BI CONNECTION



DATA MODEL + DAX



INTERACTIVE DASHBOARDS



BUSINESS INSIGHTS

This architecture demonstrates the complete transition from raw healthcare records to business decision support.

Data Preparation, SQL Analysis and BI Methodology

The first stage of the project involved preparing the raw healthcare data using Python.

Python was used for:

- Inspecting the raw datasets
- Understanding dataset structures
- Identifying relevant fields
- Preparing datasets for analysis
- Handling data-quality issues
- Standardizing fields where required
- Preparing clean datasets for database loading

This step was important because data-quality problems at the beginning of the pipeline could affect SQL calculations, relationships, KPIs, and Power BI visualizations.

After preparation, the cleaned healthcare data was organized within a SQL/MySQL database.

The database layer provided a structured environment for storing and querying the healthcare entities. This allowed the project to move from:

Raw files → Structured relational data

SQL was then used for analytical queries covering areas such as:

- Patient counts
- Encounter counts
- Procedure counts
- Encounter-class analysis
- Payer analysis
- Claim analysis
- Coverage analysis
- Aggregations
- Group-based comparisons
- Business-oriented summaries

The SQL stage provided an analytical layer before the data was consumed by Power BI and demonstrated that the project was not dependent solely on Power BI for analysis.

The structured data was subsequently connected to Power BI. Within Power BI, the project involved:

- Connecting to the analytical data
- Organizing the data model
- Creating relationships between relevant entities
- Creating measures
- Applying interactive filters
- Building dashboards
- Formatting visuals for business reporting

DAX was used to create important metrics such as:

- Total Patients
- Total Encounters
- Total Procedures
- Total Claims Cost
- Total Payer Coverage
- Average Patient Age
- Average Encounters per Patient
- Average Claim Cost
- Uncovered Claim Amount
- Procedure-to-Encounter Ratio
- Typical Visit Duration

Procedure-to-Encounter Ratio

The Services Analysis dashboard uses:

Procedure-to-Encounter Ratio = Total Procedures ÷ Total Encounters

This allows service intensity to be compared between encounter classes.

For example, the Wellness encounter class has a ratio of approximately 2.63, meaning that there are approximately 2.63 procedures for every wellness encounter when calculated across the dataset.

The project also uses Typical Visit Duration rather than relying on the mean encounter duration because the encounter data contains unusually long records that can significantly influence the arithmetic average. The median provides a more representative indication of a typical encounter duration.

Dashboard 1 – Healthcare Overview

The Healthcare Overview dashboard provides the high-level view of the hospital's overall healthcare activity.

It acts as the starting point for the analysis and establishes the overall scale of patients, procedures, encounters, claims, and payer coverage.

<u>KPI</u>	<u>Value</u>
Total Patients	974
Total Claims Cost	\$10,15,14,375.5
Total Procedures	47,701
Total Payer Coverage	\$3,10,97,506.99
Total Encounters	27,891

Visual Analysis

Total Encounters by Encounter Class

- X-axis: Encounter Class
- Y-axis: Total Encounters

Compares encounter volume across healthcare encounter classes and identifies the types responsible for the largest share of healthcare activity.

Total Claims Cost by Encounter Class

- X-axis: Total Claims Cost (\$)
- Y-axis: Encounter Class

Compares the financial burden associated with different encounter classes.

Total Encounters Trend

- X-axis: Year
- Y-axis: Total Encounters

Provides a historical view of encounter activity and helps identify changes and trends over time.

Total Claims Cost Trend

- X-axis: Year
- Y-axis: Total Claims Cost

Tracks the movement of claim costs over time.

Business Question Answered

What is the overall scale of healthcare activity, which encounter classes dominate utilization, and how have encounter volume and claim costs changed over time?

Dashboard 2 – Patient Analysis

The Patient Analysis dashboard moves from overall healthcare activity to the patient population and its utilization behavior.

Interactive Filters

- Gender
- Marital Status
- Age Group
- City
- Encounter Class

<u>KPI</u>	<u>Value</u>
Total Patients	974
Average Patient Age	74.24
Average Encounters per Patient	28.64
Average Claim Cost	\$3,639.68
Total Encounters	27,891

Visual Analysis

Patients by Encounter Class

Shows the patient distribution across different encounter classes.

Encounters per Patient by Age Group

- X-axis: Average Encounters per Patient
- Y-axis: Age Group

Highlights differences in healthcare utilization between age groups.

Patients by Payer

- X-axis: Total Patients
- Y-axis: Payer

Compares patient volumes associated with different payers.

Patient Utilization vs Claim Cost

- X-axis: Total Encounters

- Y-axis: Total Claims Cost (\$)

Explores the relationship between healthcare utilization and claim cost.

Business Question Answered

Who are the patients, how frequently are they using healthcare services, how does utilization vary across age groups, and how does utilization relate to claim cost?

Dashboard 3 – Financial Analysis

The Financial Analysis dashboard focuses on the economic side of healthcare activity, with particular emphasis on claims, payers, insurance coverage, and uncovered financial exposure.

Interactive Filters

- Gender
- Marital Status
- Age Group
- City
- Encounter Class

<u>KPI</u>	<u>Value</u>
Total Base Encounter Cost	\$32,40,421.4
Uncovered Claim Amount	\$7,04,16,868.53
Total Claims Cost	\$10,15,14,375.5
Average Claim Cost	\$3,640
Total Payer Coverage	\$3,10,97,506.99

Visual Analysis

Total Claim Cost by Payer

- X-axis: Total Claims Cost (\$)
- Y-axis: Payer

Identifies payers associated with the largest claim amounts.

Claim Cost by Coverage Status

Shows claim distribution across:

- Uncovered
- Partially Covered
- Fully Covered

Uncovered Claim Amount by Payer

- X-axis: Uncovered Claim Amount (\$)
- Y-axis: Payer

Highlights payers associated with significant uncovered financial amounts.

Claim Burden and Insurance Coverage by Encounter Class

Compares:

- Total Claims Cost
- Payer Coverage

across encounter classes.

Business Question Answered

What is the financial burden of healthcare activity, which payers contribute most to claims, and where does the greatest uncovered financial exposure occur?

Dashboard 4 – Services Analysis

The Services Analysis dashboard focuses on healthcare service delivery, particularly encounters, procedures, procedure volume, and service intensity.

Interactive Filters

- Gender
- Marital Status
- Age Group
- City
- Encounter Class

<u>KPI</u>	<u>Value</u>
Total Patients	974
Total Encounter Classes	6
Total Procedures	47,701
Total Encounters	27,891
Typical Visit Duration	15 min

The Typical Visit Duration uses the median encounter duration to provide a representative measure that is less influenced by unusually long encounters.

Visual Analysis

Claim Cost by Encounter Class

- X-axis: Encounter Class
- Y-axis: Claim Cost

Shows how claim activity varies between encounter classes.

Top 5 Procedures by Volume

Displays the five most frequently performed procedures in the dataset and identifies the healthcare services with the highest procedure volume.

Procedure-to-Encounter Ratio by Encounter Class

- X-axis: Encounter Class
- Y-axis: Procedure-to-Encounter Ratio

Measures the relative number of procedures associated with each encounter class.

For example:

Wellness = 2.63

means approximately 2.63 procedures per wellness encounter across the dataset.

This is a ratio across the dataset and does not mean that every individual wellness visit contains exactly 2.63 procedures.

Business Question Answered

Which healthcare procedures are performed most frequently, how does service intensity vary across encounter classes, and what is a typical healthcare visit duration?

Key Business Insights

The analysis contains 974 patients and 27,891 encounters, indicating substantial healthcare utilization across the population.

The age-group analysis shows that the 65+ population has considerably higher encounter utilization compared with the younger groups displayed in the dashboard. This indicates that patient age is an important factor when examining healthcare utilization.

The financial analysis identifies:

- Total Claims Cost: \$10,15,14,375.5
- Total Payer Coverage: \$3,10,97,506.99
- Uncovered Claim Amount: \$7,04,16,868.53

This highlights a significant uncovered component within the overall claim amount. Payer-level and coverage-status analysis helps identify where this financial exposure is concentrated.

The dataset contains:

47,701 procedures across 27,891 encounters.

Procedure volume is not evenly distributed across encounter classes. The Procedure-to-Encounter Ratio provides an additional service-intensity perspective.

For example:

Wellness → 2.63 procedures per encounter

while other encounter classes have substantially lower ratios.

Therefore, the project does not only identify the volume of procedures, but also examines procedure intensity relative to encounter volume.

Overall, the dashboards provide complementary views of the hospital's:

- Patient population
- Healthcare utilization
- Encounter activity
- Procedure activity
- Claim costs
- Payer coverage
- Uncovered financial exposure
- Service intensity
- Historical trends

Conclusion

CareScope successfully implements an end-to-end healthcare data analytics workflow:

Python → SQL/MySQL → Power BI → Business Insights

The project demonstrates the complete process of:

Data Preparation → Database Management → SQL Analysis → BI Modeling →
Visualization → Business Interpretation

The final four dashboards provide complementary analytical perspectives:

Dashboard	Primary Focus
Healthcare Overview	Overall healthcare activity
Patient Analysis	Patient utilization
Financial Analysis	Claims and payer coverage
Services Analysis	Procedures and service intensity

The project provides a centralized analytical solution for understanding healthcare activity, patient utilization, encounter patterns, procedure volume, service intensity, claims, payer contribution, insurance coverage, uncovered financial exposure, and historical healthcare trends.

More importantly, it demonstrates a complete Data Analyst workflow, rather than a dashboard-only project.

The integration of Python for data preparation, SQL/MySQL for database-driven analysis, and Power BI for interactive Business Intelligence reporting demonstrates how multiple technologies can be combined to transform raw healthcare data into a practical analytical solution.

CareScope provides a unified view of hospital operations by connecting patient, encounter, procedure, payer, claim, and coverage information, enabling stakeholders to understand healthcare utilization, service delivery, and financial performance through data-driven analysis.